Operational non-split Richelot transfer over F17

2026-11-11 · Chained ECDLP Transfers · fixed full prime-order19 source

Unsigned/unverified standing. The actual determinant11 correspondence and dual operate on every285 rational class. This is same-field arithmetic, without a DLP solver or demonstrated computational advantage. PCC383894 and bd303e bind development/full replay. [pcc:unverified]

Actual correspondence and balanced descent

The fixed quadratic factors are a Frobenius-stable unordered triple over $F17^3$. Source support has degree≤2 and splits over $F17^2$. Their compositum $F17^6$ suffices: retain each target fiber as a quadratic Mumford pair, without choosing target roots or an $F17^{12}$ splitting. Raw target ordinate w=3Y fixes the monic normalization and dual scaling. [pcc:unverified]

For source finite degree m and infinity count n, include the full balanced sum $\Sigma\psi(P)+(2n+m-2)\psi(\infty+)$. Individual fiber classes may retain extension coefficients; completed outputs must descend to F17 and pass the separate prime-field validator. Degree drops, multiplicity and infinity contributions are explicit. [pcc:unverified]

All285 source and target classes pass both dual norms[2]; paired additivity and285 distinct images pass. The isogeny has geometric degree4 while its rational kernel has1 element, consistent with odd class-group order285. All19 elliptic scalar relations survive; reversed-chain/source norm gives[4], recovered by5 modulo19. These recover points, not unknown DLP scalars. [pcc:unverified]

The earlier complete PGL2 audit distinguishes source/target curves overF17; the second nondegenerate kernel gives a reflection-equivalent target. No unpolarized-Jacobian nonisomorphism is inferred. Retain both equivalence and degenerate cases rather than counting more destinations. [pcc:unverified]

Independent arithmetic, algebra and remaining limits

Crypto’s pre-existing flat degree6 Fpk/Rabin engine independently replays 128 tower-field products, inverses and Frobenius powers. It imports none of the new field/map code. The explicit bridge is gamma=T+s, s=(gamma³+11gamma+2)/(3gamma²+5), with flat modulus gamma⁶+12gamma⁴+4gamma³+14gamma²+10gamma+14. PCC167595 binds the TSV and passing receipt. The engine is unchanged; only a unit test and fixture were added. [pcc:unverified]

That validates bounded extension arithmetic, not independently the entire Jacobian map. Both balanced engines use Cantor; harnesses and receipts are producer-authored, unsigned evidence. No signed scientific verdict is claimed. [pcc:unverified]

Four generic core Lean lemmas pass a7-declaration gate, crypto PCCf7eb4f. The fundamental identity for arbitrary quadratics is ΣGi(x)[G’jGk−GjG’k](z)=−δ(x−z)². The other certificates establish the cleared target-curve equation under the correspondence, balanced infinity coefficient and inverse5 recovery under19-torsion. Actual finite fields, projective/Picard extension and global dual construction remain written inputs from the primary sources. [pcc:unverified]

Reproduction and retained attempts

Consumer check.sh passes exact source guards, formatting/strict Clippy, native25/reference43 tests and1 pre-existing ignored benchmark. Receipt nonsplit-map-pinned-01.txt is retained. Root command: cargo test –release –bin icprog chained_nonsplit_compositum_witnesses – –nocapture. The minimal sparse-checkout script automatically includes the committed TSV. The first map run passed with an import warning, subsequently cleaned; the first new Lean module needed explicit variable-scalar rewrites. Failed proof/format attempts and passing receipts remain separate. [pcc:unverified]

Full account: NONSPLIT_MAP.md; source src/nonsplit_map.rs. Primary references: Kunzweiler,DCC92(2024),§§3.1–3.2,Propositions7–8; Galbraith corrected ch.8§8.3/ch.10§10.4.2. Upstream source receipt binds bytes.

Next bounded task

CT15 predeclares shift2 of the existing E109/F101 source after this pipeline validation. The shifted cover family is prior art, and the unshifted6-cycle negative remains retained. No shifted map is yet accepted. Larger fields, catalog/production integration and independent signed validation remain open. [pcc:unverified]

No benchmark, cost ratio, solver comparison or paid job ran. S3/live-budget, KB/vault and crypto frozen-ledger/archive blockers remain. No prime-field attack or cross-prime descent is established. Canonical graphs remain current: no signed scientific or measured performance edge has been promoted.